

Chapter 14 - Firewall, System Settings & Network Troubleshooting

1. Firewall (Very Important Security Concept)

◆ What is Firewall?

A **Firewall** is a **hardware or software security system** that **filters incoming and outgoing network traffic** based on rules.

 It acts like a **security guard / traffic police** 

Real-Life Example:

- If documents are valid → Allow 
- If suspicious → Block 

Same as:

- Firewall checks:
 - IP address 
 - Port 
 - Application 
 - URL 
-

◆ Types of Firewall

Software Firewall

- Installed inside your system
 - Example: Windows Defender Firewall
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2 Hardware Firewall 🌐

- Physical device
- Installed before router/network

👉 Example:

Company network security device

◆ What Can Firewall Do?

- ✓ Block apps
 - ✓ Block websites
 - ✓ Block ports
 - ✓ Allow trusted traffic
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◆ Practice Tool

👉 Use **FortiGate Firewall (Free VM version)**

- Download OS
 - Run in VMware
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◆ Windows Firewall Settings

📌 Control Panel → Windows Defender Firewall

🔧 **Important Options:**

✓ Allow an App Through Firewall

- Allow/block applications

👉 Example:

Block Chrome → No internet in Chrome

✓ Notification Settings

- Show alert when app blocked 🔔

⚠ Important Rule:

- 👉 Only **ONE Firewall + ONE Antivirus** should run at a time

🔧 Troubleshooting:

Problem	Solution
Internet not working in one app	Check firewall rules
App blocked	Allow through firewall
No notification	Enable alerts

⚙ 2. System Settings (Display, Sound, Recovery)

💻 Display Settings

- Brightness 🌞
 - Night Light 🌙
 - Resolution 📺
 - Orientation 🔄
 - Multiple Displays 💻
-

Issue: Resolution Not Changing

Cause:

- Display driver issue

Solution:

- Update GPU/Display driver
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Sound Settings

Common Issue:

User joins meeting (Teams/Zoom) → Cannot hear sound

Fix:

1 Check Windows sound settings

2 Check app sound settings

3 Select correct:

- Speaker 
 - Headphone 
 - Bluetooth device
-

Advanced Case:

 Driver working

 Settings correct

 Still no sound?

 **Hardware issue (Speaker damaged)**

Real-Life Example:

Speaker detected but no sound → Replace hardware

Notifications

- Do Not Disturb ON → No notifications ❌
-

Recovery (Advanced Startup)

◆ Use Case:

- System slow after update
-

Solution:

- 1 Settings → Recovery
 - 2 Advanced Startup
 - 3 Enter recovery mode
 - 4 Uninstall updates
-

3. Bluetooth & Devices

◆ Common Issue:

Bluetooth option missing

Fix:

- Check driver
 - Check service
-

◆ **Features:**

- Add device 📱
- Configure:
 - Mouse 🖱️
 - Keyboard ⌨️
 - Camera 📷
 - Touchpad

 4. Network & Internet (VERY IMPORTANT 🔥)

 Wi-Fi Settings

- Shows:
 - Connected networks
 - Previously connected networks

 Error: “Can’t Connect to This Network”

 Reason:

- Wrong saved password
 - Corrupted network profile
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Solution 1: Settings Method

- 1 Go to:
Settings → Network & Internet → Wi-Fi
 - 2 Click:
 Manage Known Networks
 - 3 Select network
 - 4 Click **Forget**
 - 5 Reconnect with correct password
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Solution 2: Registry Method (Advanced)

 Sometimes password is still stored internally

Path:

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows
NT\CurrentVersion\NetworkList\Profiles

Steps:

- 1 Open Run → regedit
 - 2 Go to path above
 - 3 Open each profile
 - 4 Find Wi-Fi name (ProfileName)
 - 5 Delete that profile
 - 6 Restart system
 - 7 Reconnect Wi-Fi
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Real-Life Example:

User forgot network but still error → Registry cleanup fixes it

Ethernet Settings

◆ Network Profile Types:

Type	Use Case
Public 	Secure (Recommended)
Private 	For file sharing

⚠ Important:

👉 Use **Public Network** in:

- Airport 
- Cafe 

👉 Use **Private Network** in:

- Office 
 - Home 
-

Issue:

File sharing not working

👉 Cause:

- Network set to Public

👉 Fix:

- Change to Private
-



Network Reset

◆ When to Use?

- Internet slow
 - All fixes failed
-



Steps:

Settings → Network → Advanced → Network Reset



Other Network Features

- VPN 
 - Mobile Hotspot 
 - Airplane Mode 
 - Proxy 
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5. Personalization

- Background 
 - Themes 
 - Colors 
 - Lock Screen 
 - Taskbar 
 - Fonts 
-

6. Apps & Features

◆ Installed Apps:

- View all apps
-

◆ Default Apps:

- Set:
 - PDF → Adobe
 - Browser → Chrome
-

◆ Advanced Settings:

- Uninstall updates
 - Repair apps
-

7. Accounts Section

Includes:

- Your Info 
 - Sign-in Options 
 - Linked Devices 
 - Work/School Account 
 - Other Users 
 - Windows Backup 
-

8. Time & Language

- Date & Time 
 - Language 
 - Region 
 - Keyboard 
-

9. Privacy & Security

◆ Includes:

- Windows Security 
 - Device Encryption 
 - Find My Device 
-

◆ App Permissions:

- Location 
 - Camera 
 - Microphone 
 - Notifications 
 - Files & Documents 
-

10. Windows Update

◆ Features:

- Install updates
 - Pause updates
 - View update history
-

⚠ Windows Insider Program

👉 Early access to new features

✗ Not recommended (Buggy)

💡 Real-Life Example:

User installs insider build → System unstable

⚡ Final IT Support Mindset (Important)

🔥 Always Diagnose Like This:

1 Is it user issue or system issue?

2 Check:

- Driver
- Settings
- Network
- Hardware

3 Fix root cause (not temporary)

Quick Troubleshooting Table

Problem	Solution
Wi-Fi not connecting	Forget + Registry delete
No sound	Check device + hardware
Slow internet	Network reset
App no internet	Check firewall
Bluetooth missing	Driver/service fix
File sharing not working	Change to Private network

Final Summary

- Firewall filters traffic like security guard 🔥
 - Display/sound issues mostly driver-related 🎧
 - Wi-Fi errors fixed via settings + registry 🌐
 - Network profile impacts security 🔒
 - System settings knowledge = strong troubleshooting ⚡
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Q & A

FIREWALL – INTERVIEW QUESTIONS & ANSWERS

 **Q1: What is a Firewall?**

 **Answer:**

A Firewall is a security system (hardware or software) that monitors and filters incoming and outgoing network traffic based on predefined rules.

 **Simple Explanation:**

Firewall works like a **security guard**  that checks every request before allowing or blocking it.

 **Q2: What does a Firewall check?**

 **Answer:**

Firewall filters traffic based on:

- IP Address 
 - Port Number 
 - Application 
 - URL / Website 
-

 **Q3: Types of Firewall?**

 **Answer:**

1 Software Firewall 

- Installed inside OS
- Example: Windows Defender Firewall

2 Hardware Firewall 🌐

- Physical device
 - Used in companies
-

? Q4: What can a firewall do?

✓ Answer:

- ✓ Block applications
 - ✓ Block websites
 - ✓ Block ports
 - ✓ Allow trusted traffic
-

? Q5: Why should only one firewall be used?

✓ Answer:

👉 Multiple firewalls can:

- Conflict with each other ✖
 - Block valid traffic
 - Slow down system
-

? Q6: How to allow an app through firewall?

✓ Answer:

Steps:

1. Open Control Panel
2. Windows Defender Firewall
3. Click:
👉 Allow an app through firewall
4. Select app → Allow

? **Q7: Real scenario – App not getting internet. What will you do?**

✓ **Answer:**

Step-by-step:

1. Check firewall rules 🔥
2. Check if app is blocked
3. Allow app in firewall
4. Disable firewall temporarily (for testing)

⚙️ **SYSTEM SETTINGS – INTERVIEW QUESTIONS**

? **Q8: What are display settings used for?**

✓ **Answer:**

- Brightness 🌞
- Resolution 📺
- Orientation 🔄
- Multiple displays 🖥️

? **Q9: Resolution not changing. What is the issue?**

✓ **Answer:**

👉 **Cause:**

- Display driver issue

👉 **Solution:**

- Update graphics driver
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? Q10: No sound in system. How will you troubleshoot?

✓ Answer:

Step-by-step:

1. Check sound settings 🔊
2. Check output device (speaker/headphone)
3. Check application settings (Teams/Zoom)
4. Update audio driver
5. Check hardware

🎯 If everything correct → hardware issue

? Q11: Notifications not showing?

✓ Answer:

👉 Cause:

- Do Not Disturb ON

👉 Fix:

- Turn OFF Do Not Disturb
-

? Q12: System slow after update. What will you do?

✓ Answer:

Steps:

1. Go to Recovery
 2. Advanced Startup
 3. Uninstall updates
-

BLUETOOTH & DEVICES – INTERVIEW QUESTIONS

 **Q13: Bluetooth option missing. What is the issue?**

 **Answer:**

 **Cause:**

- Driver missing
- Service disabled

 **Fix:**

- Install Bluetooth driver
 - Enable Bluetooth service
-

 **Q14: How to connect Bluetooth device?**

 **Answer:**

Steps:

1. Settings → Bluetooth
 2. Turn ON
 3. Click Add Device
 4. Pair device
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NETWORK & INTERNET – MOST IMPORTANT

 **Q15: “Can’t connect to this network” error. How to fix?**

 **Answer:**

Method 1:

1. Go to Wi-Fi settings
2. Manage known networks

3. Click Forget
 4. Reconnect
-

Method 2 (Advanced):

1. Open Registry
2. Go to:

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows
NT\CurrentVersion\NetworkList\Profiles

3. Delete Wi-Fi profile
 4. Restart
-

? Q16: Difference between Public and Private network?

✓ Answer:

Type	Use
Public 🌐	Secure (Cafe, Airport)
Private 🏠	File sharing (Home/Office)

? Q17: File sharing not working. Why?

✓ Answer:

👉 Cause:

- Network set to Public

👉 Fix:

- Change to Private
-

? Q18: When to use Network Reset?

✓ Answer:

👉 Use when:

- Internet not working
 - All troubleshooting failed
-

? Q19: Slow internet issue. What will you check?

✓ Answer:

1. Check network usage
 2. Restart router
 3. Run network reset
 4. Check driver
 5. Check background apps
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? Q20: VPN, Proxy, Hotspot – what are they?

✓ Answer:

- VPN 🗝️ → Secure connection
 - Proxy 🌐 → Middle server
 - Hotspot 📶 → Share internet
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APPS & SYSTEM SETTINGS QUESTIONS

? Q21: Default apps meaning?

✓ Answer:

Apps used by default to open files.

👉 Example:

- PDF → Adobe
- Browser → Chrome

? Q22: How to uninstall app?

✓ Answer:

- Settings → Apps → Installed Apps
- Click uninstall

PRIVACY & SECURITY QUESTIONS

? Q23: What permissions can be controlled?

✓ Answer:

- Camera 
 - Microphone 
 - Location 
 - Files 
-

? Q24: What is Windows Update?

✓ Answer:

System to install updates for:

- Security 🛡️
 - Performance ⚡
-

? Q25: Why avoid Insider Program?

✓ Answer:

👉 Because:

- Buggy ✖️
 - Unstable system
-

💡 REAL JOB SCENARIO QUESTIONS (VERY IMPORTANT)

? Q26: User says Chrome not working but internet works. What will you do?

✓ Answer:

1. Check firewall 🔥
 2. Check proxy settings
 3. Reset browser
 4. Check extensions
-

? Q27: Wi-Fi connected but no internet

✓ Answer:

1. Check IP address
2. Restart router

3. Run network reset
 4. Check DNS
-

? Q28: Bluetooth device connected but not working

✓ Answer:

1. Remove device
 2. Reconnect
 3. Update driver
 4. Check compatibility
-

? Q29: No sound in meeting app

✓ Answer:

1. Check app audio settings
 2. Check Windows output device
 3. Restart app
 4. Update driver
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? Q30: System slow

✓ Answer:

1. Check Task Manager
 2. Disable startup apps
 3. Update drivers
 4. Scan for malware
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ADVANCED INTERVIEW QUESTIONS

 **Q31: How firewall helps in security?**

 **Answer:**

- Blocks unauthorized access
 - Prevents hacking attempts
 - Filters malicious traffic
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 **Q32: What happens if firewall is disabled?**

 **Answer:**

- System becomes vulnerable
 - Malware can enter easily
 - Network attacks possible
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